

# 114-3□□□□□□\_Advanced 1D Kinematics\_□□



## Useful Formulas for kinematics

(1)  $v = v_0 + at$

(2)  $\Delta x = v_0 t + \frac{1}{2}at^2$

(3)  $v^2 = v_0^2 + 2a\Delta x$

Q1: A bicycle's velocity increases steadily from 4 m/s to 16 m/s in 6 s.

(a) What is its average acceleration?

(b) Draw the Velocity-Time (v-t) graph for this bicycle. Calculate the displacement from  $t = 0$  to 6 s from v-t graph.

Q2: A car accelerates from rest at a constant rate of 3 m/s<sup>2</sup>.

(a) How much time does it take for the car to reach a final velocity of 15 m/s?

(b) Calculate the distance the car traveled during this time.

Q3: A train is traveling at 30 m/s. The engineer hits the brakes, and the train comes to a complete stop ( $v = 0$  m/s) over a distance of 90 m.

(a) Calculate the train's constant acceleration.

(b) How many seconds it took for the train to stop.

Q4: A toy rocket is launched straight up into the air. Gravity acts on it with  $a = -10$  m/s<sup>2</sup>. It reaches a maximum height of 20 m before falling back down. At its maximum height, its instantaneous velocity is exactly 0 m/s.

- (a) Calculate its initial launch velocity.
- (b) Draw the Velocity-Time (v-t) graph for the rocket from launch until it reaches its highest point. Use the graph to find out how many seconds it took to reach the top.

Q5: A cyclist rides in a two-stage motion.

- (a) Stage 1: She starts from rest and accelerates at  $3 \text{ m/s}^2$  for 4 s. What is her speed at the 4-second mark? How far did she ride during Stage 1?
- (b) Stage 2: She then maintains that top speed (constant velocity) for another 5 s. How far did she ride during Stage 2?
- (c) Draw a v-t graph for this entire 9-second ride and calculate the total displacement via the graph.

Q6: Car A is driving at  $30 \text{ m/s}$ . Car B is  $75 \text{ m}$  directly ahead of Car A, driving in the same direction at a constant velocity of  $10 \text{ m/s}$ . The driver of Car A hits the brakes, creating a constant deceleration of  $-2 \text{ m/s}^2$ .

- (a) The danger of a collision only exists while Car A is moving faster than Car B. Calculate how many seconds it takes for Car A to slow down matching Car B's speed.
- (b) Using the time you found in (a), calculate the displacement of Car A and the displacement of Car B during this specific time period.
- (c) Assume Car A started at position  $x_A = 0 \text{ m}$ . This means Car B started at  $x_B = 80 \text{ m}$ . What are their final positions at time t? Based on these positions, did Car A crash into Car B?
- (d) When they exactly crush to each other, which physical quantity of two cars will be the same? List the equation and solve for the exact second the crash happens.